

Life Simulator

Department of Software Engineering

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Section:

AB

Course Title:

Programming Fundamentals

Course Code:

SE-101

Submitted to:

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Date Submitted:

12/30/2025

Code

```
#include<iostream>
#include<string>
using namespace std;

int money = 0;
int bankCash = 25000;

struct hotel {
   string name;
   float rating;
   string bFood;
   string location;
};

struct ride {
   string name;
   int id;
   float price;
};

struct veg {
   string name;
   float price;
};

void bankVisit(int bankChoice) {
   int choice;
   float cash;
```

Code

```
  int changeChoice;


  if (bankChoice == 1) {
    cout << "ATM: How much money do you want to withdraw?" << endl;
    do {
      cin >> cash;
      if (cash < bankCash) {
        money = money + cash;
         bankCash = bankCash - cash;
         break;
      } else {
         cout << "ATM: Try again! You are exceeding the maximum limit" << endl;
      }
    } while (true);
  } else {
    cout << "Current Cash in Hand: " << money << endl;
    if (money > 1000) {
        cout<<"Should I keep some change for myself?"<<endl;
        cin>>changeChoice;
        if(changeChoice==1){
        int keptAmount = money * 0.10;
      int depositAmount = money - keptAmount;
      bankCash = bankCash + depositAmount;
      money = keptAmount;
      cout << "Deposited Amount: " << depositAmount << endl;
} else {
      bankCash = bankCash + money;
      money = 0;
```

Code

```
        cout << "Deposited all cash to the bank." << endl;
      }
      } else {
        bankCash = bankCash + money;
        money = 0;
        cout << "Deposited all cash to the bank." << endl;
      }
      cout << "New Bank Balance: " << bankCash << endl;
}
}

void vegShop(veg vegMenu[5]) {
   int choice;
   int amount;
   int count = 0;
   veg bill[100];
   int qty[100];
   int totBill = 0;

   cout << "Item\t\tPrice" << endl;
   for (int i = 0; i < 5; i++) {
      cout << i + 1 << ". " << vegMenu[i].name << "\t" << vegMenu[i].price << endl;
   }
   cout << "Shopkeeper: Choose item (0 to exit)" << endl;

    do {
      cin >> choice;
      if (choice > 0 && choice <= 5) {
```

Code

```
      int i = choice - 1;
      cout << "Shopkeeper: How much do you want to buy sir?" << endl;
      cin >> amount;

      if (money >= vegMenu[i].price * amount) {
         bill[count] = vegMenu[i];
         qty[count] = amount;
         totBill = totBill + (vegMenu[i].price * amount);
         count++;
         cout<<"Shopkeeper: Want anything else sir?"<<endl;
      } else {
         cout << "Me: I dont have enough balance. Lets go to the bank again" << endl;
         bankVisit(1);
         cout << "Me: Lets try shopping again" << endl;
      }
   } else if (choice == 0) {
      if (count == 0) {
         cout << "Me: I need to buy at least 1 item" << endl;
      } else {
         break;
      }
   } else {
      cout << "Shopkeeper: Sorry sir! We don't have it available at the moment" << endl;
   }
} while (choice != 0);

cout << "Item\t\tPrice\t\tQuantity" << endl;
```

Code

```
    for (int i = 0; i < count; i++) {
        cout << i + 1 << ". " << bill[i].name << "\t" << bill[i].price << "\t\t" << qty[i] << endl;
    }
    cout << "Total Bill: " << totBill << endl;
    money = money - totBill;
    cout << "Shopkeeper: Thanks for shopping here sir." << endl;
}

void park(bool isFriend) {
    int rideChoice;
    ride bill[10];
    int totalBill = 0;
    int count = 0;
    cout << "Ticket Machine: Welcome to Fatima Jinnah Park! Please Enter the Ride ID you want to choose (Press 0 to Exit) "<<endl;

    ride rides[4] = {
        {"Slide  ", 1, 20},
        {"See-Saw", 2, 40},
        {"Ping   ", 3, 60},
        {"Cars   ", 4, 100},
    };

    cout << "ID\tName\t\tPrice" << endl;
    for (int i = 0; i < 4; i++) {
        cout << rides[i].id << "\t" << rides[i].name << "\t\t" << rides[i].price << endl;
    }
```

Code

```
  do {
    cin >> rideChoice;

    if (rideChoice == 0) {
      break;
    }

    if (rideChoice > 0 && rideChoice <= 4) {
      int i = rideChoice - 1;
      int actualPrice = rides[i].price;

      if (isFriend == true) {
        actualPrice = actualPrice * 2;
        cout << "Ticket Machine: For 2 people, The price of tickets would be: " <<
actualPrice << endl;
      }

      if (money >= actualPrice) {
        money = money - actualPrice;
        bill[count] = rides[i];
        bill[count].price = actualPrice;
        totalBill = totalBill + actualPrice;
        count = count + 1;
      } else {
        cout << "Me: I dont have enough balance. Lets go to the bank again" << endl;
        bankVisit(1);
        cout << "Me: Lets try buying the ticket again" << endl;
      }
```

Code

```
    } else {
        cout << "Ticket Machine: Invalid Ride ID." << endl;
    }

  } while (count < 4);

  cout << "Item\t\tPrice" << endl;
  for (int i = 0; i < count; i++) {
    cout << i + 1 << ". " << bill[i].name << "\t" << bill[i].price << endl;
  }
  cout << "Ticket Machine: Your Total Bill is: " << totalBill << endl;
  cout << "Ticket Machine: Thanks for visiting here sir." << endl;
}

void hotels(bool isFriend) {
  int hotChoice;
  int fChoice;
  veg bill[10];
  int count = 0;
  int totBill = 0;

  hotel H[4] = {
    {"Sialvi Madni Restaraunt       ", 4.5, "Chicken Karahi ", "Lalamusa"},
    {"Bismillah Burgers & Shawarma   ", 4.3, "Shaami Burger  ", "Lalamusa"},
    {"Karachi Student Chicken Biryani", 4.6, "Chicken Biryani", "Gujrat"},
    {"Kentucky Fried Chicken (KFC)   ", 4.7, "Chicken Wings  ", "Kharian"}
  };
```

Code

```
   veg hotelMenus[4][4] = {

      { {"Chicken Karahi ", 900}, {"Mutton Karahi  ", 2500}, {"Chicken Fried Rice", 700},
{"Naan/Roti Basket   ", 200} },

      { {"Shaami Burger  ", 160}, {"Zinger Burger  ", 350}, {"Chicken Shawarma(L)", 250},
{"Chicken Shawarma(S)", 190} },

      { {"Chicken Biryani", 160}, {"Simple Biryani ", 110}, {"Shami Tikki & Salad", 50}, {"Soft
Drink (250ML)  ", 70} },

      { {"Chicken Wings  ", 500}, {"Chicken & Chips", 400}, {"Zingeratha        ", 390},
{"Krunch Burger     ", 310} }

   };


   cout << "\t\t\t\tSELECT A HOTEL" << endl;

   cout << "\tName" << "\t\t\t\t" << "\tRatings"<< "\t\t" << "Best Food"<< "\t" <<
"Location"<< endl;

   for (int i = 0; i < 4; i++) {

      cout << i + 1 << ". " << H[i].name << "\t\t" << H[i].rating<< "\t\t" << H[i].bFood<< "\t\t"
<< H[i].location << endl;

   }

   cin >> hotChoice;


   if (hotChoice >= 1 && hotChoice <= 4) {

      int hIndex = hotChoice - 1;

      cout << "Waiter: Welcome to " << H[hIndex].name << "! sir, please take a look at our
menu" << endl;

      cout << "\t\t\tMENU" << endl;

      cout << "\tItem" << "\t\t\t\tPrice" <<endl;

      for (int i = 0; i < 4; i++) {

         cout << i + 1 << ". " << hotelMenus[hIndex][i].name

            << "\t\t\t" << hotelMenus[hIndex][i].price << endl;

      }
```

Code

```
    do {
      cout << "(Enter Item Number to Order (0 to exit)): ";
      cin >> fChoice;

      if (fChoice > 0 && fChoice <= 4) {
        int itemIndex = fChoice - 1;
        veg selectedItem = hotelMenus[hIndex][itemIndex];

        int actualPrice = selectedItem.price;

        if (isFriend == true) {
          actualPrice = actualPrice * 2;
          cout << "Waiter: Ok sir! That would be for 2 people right? " << actualPrice <<
endl;
          cout<<"Me: Yes"<<endl;
        }

        if (money >= actualPrice) {
          money = money - actualPrice;
          selectedItem.price = actualPrice;
          bill[count] = selectedItem;
          totBill = totBill + actualPrice;
          count++;
        } else {
          cout << "Me: Not enough cash! Going to bank..." << endl;
          bankVisit(1);
          cout << "Me: Alright I am Back with money!" << endl;
        }
```

Code

```
        } else if (fChoice != 0) {
            cout << "Waiter: Sorry Sir, We don't have what you are asking for available at the moment'" << endl;
        }

    } while (fChoice != 0 && count < 10);

    cout << " Cashier: Sir your Total Bill would be: " << totBill << endl;
  } else {
    cout << "Me: I don't think I know this hotel" << endl;
  }
}

void hospital() {
cout << "Me: I am feeling extremley nauseous, doctor. I went to a hotel and after that I went to a park." << endl;
  cout << "Doctor: No need to worry, A few meds and you will be ok" << endl;
  cout << "Me: I am feeling much better now" << endl;
  cout << "Doctor: Take care next time" << endl;
  cout<<"Reception: Sir your bill would be 2000"<<endl;
  cout<<"Me: Alright here you go"<<endl;
  money = money - 2000;
}

int main() {
  cout << "Me: I need to go out today. But first, I need some cash from the bank." << endl;
  bankVisit(1);
```

Code

```
  cout << "Me: Alright Now, Lets proceed to shop for some vegetables" << endl;
  cout << "Shopkeeper: Welcome to Our Shop sir. Please, take a look at our menu" << endl;
  veg vegMenu[5] = {
    {"Potato", 120},
    {"Tomato", 300},
    {"Onion", 500},
    {"Garlic", 700},
    {"Ginger", 800}
  };

  vegShop(vegMenu);

  veg frtMenu[5] = {
    {"Apple", 120},
    {"Mango", 300},
    {"Grapes", 500},
    {"Orange", 700},
    {"Banana", 800}
  };
  cout << "Me: Alright Now, Lets proceed to shop for some fruits" << endl;
  vegShop(frtMenu);

  int frnChoice;
  int parkHotChoice;
  int extChoice;
  bool isFriend;
```

Code

```
  cout << "Mateen: Hey Where you going?" << endl;
  cout << "Me: Hey there!" << endl;
  cout << "(Press 1 to say Park and 2 to say Hotel and anything else to shrug him off)" << endl;
  cin >> frnChoice;

  if (frnChoice == 1) {
    cout << "Me: I am going to the park. Lets go together." << endl;
    parkHotChoice = 1;
    isFriend = true;
  } else if (frnChoice == 2) {
    cout << "Me: I am going to a hotel. lets go together" << endl;
    parkHotChoice = 2;
    isFriend = true;
  } else {
    cout << "Me: Nowhere bro! I am just headed home" << endl;
    cout << "(Finally, I can go alone. But where should I go first)" << endl;
    isFriend = false;
    do {
      cout << "1 for Park and 2 for hotel" << endl;
      cin >> parkHotChoice;
      if (parkHotChoice < 1 || parkHotChoice > 2) {
        cout << "Me: I dont know where that place really is" << endl;
      }
    } while (parkHotChoice < 1 || parkHotChoice > 2);
  }

  if (parkHotChoice == 1) {
```

Code

```
    park(isFriend);
    cout << "Should I go to the hotel? (1 for Yes 2 for No)" << endl;
    do
    {
       cin>> extChoice;
    if(extChoice == 1) {
       hotels(isFriend);

    } else if (extChoice == 2) {
       cout << "Nah! lets go home" << endl;
       bankVisit(2);
       cout << "I am back Home!!" << endl;
    } else {
       cout << "What does that even mean?" << endl;
    }
    } while (extChoice < 1 || extChoice > 2);

} else if (parkHotChoice == 2) {
    hotels(isFriend);
    cout << "Should I go to the park? (1 for Yes 2 to say No)" << endl;
    do {
       cin >> extChoice;
       if (extChoice == 1) {
          park(isFriend);
          cout << "(I am feeling very nauseous. I have to visit the hospital.)" << endl;
          hospital();
          bankVisit(2);
          cout << "I am back Home!!" << endl;
```

Code

```
        } else if (extChoice == 2) {
            cout << "Nah! lets go home" << endl;
            bankVisit(2);
            cout << "I am back Home!!" << endl;
        } else {
            cout << "What does that even mean?" << endl;
        }
    } while (extChoice < 1 || extChoice > 2);
  }
  cout<<"Me: It's been a long day! Now, I am going to sleep. Good Night!!!"<<endl;
  return 0;
}
```